

Shantanu Amit More

samore@clemson.edu | +1 864 325 0745 | [LinkedIn](#) | [Portfolio](#)

Availability – January 2027 to August 2027

Education

Clemson University | GPA: 4.0/4.0

August 2025 – August 2027

Master of Science in Automotive Engineering

College of Engineering, Pune (COEP) | GPA: 8.1/10

August 2019 – August 2023

Bachelor of Technology, Production Engineering and Industrial Management

Engineering Achievements

Hybrid Powertrain Controls: Designed **supervisory control logic** in Stateflow that selects torque path and propulsion source in a series-parallel hybrid powertrain. Demonstrating a 21 MPG (43 to 64) improvement in simulated fuel economy over a conventional ICE baseline on a real-world recorded drive cycle.

Optimal Torque Allocation: Tuned and validated a Model Predictive Controller using Simulink-CarSim co-simulation, reducing peak vehicle yaw rate by **83% (77 deg/s to 13 deg/s)** and side slip angle by 85% on low-friction surface.

Sensor Fusion and Estimation: Reduced ultrasonic sensor data convergence time **by 51% (from 1.96 s to 960 ms)** by implementing sensor-fusion with sequential Kalman-filters and optimizing the sampling rate to mitigate signal interference.

Brake system validation: Sized brake system components for a reverse-trike three-wheeler, successfully attaining the braking torque requirements on an 18% gradient.

Experience

Powertrain Engineer | Deep Orange 18

January 2026 – Present

- Formulated a state-space bicycle model for yaw-rate tracking active steering controller on Simulink. Tuned PID gains to achieve a damping ratio of 0.8 and a closed-loop natural frequency of 2 Hz.
- Developed a two-stage hierarchical lateral vehicle stability controller, with upper-level Mamdani Fuzzy logic and lower-level Model-predictive controller. Enforced torque-rate, slip-ratio and actuator constraints to track desired vehicle yaw-rate and sideslip angle.
- Validated both controllers through MIL simulations, using CarSim as the vehicle plant across various lane-change maneuvers and sinusoidal steering inputs.
- Established a control-loop between vehicle sensors and actuators over CAN communication using a New Eagle Raptor ECU.

Research Assistant | Clemson University

August 2025 – May 2026

- Extracted cell internal resistance and capacitance from HPPC tests and simulated a second order (2-RC) equivalent-circuit model of the cell in MATLAB using the real-world cell parameters.
- Developed an algorithm to estimate cell state of charge based on Kalman Filter.

Transmission System Lead | Team Octane Racing Electric | Formula SAE, COEP

July 2021 – May 2023

- Led a five-member team and collaborated cross-functionally to design and integrate a custom limited-slip differential, reducing the vehicle lap-time by 1.1 seconds.
- Built a differential-equation based model of a 32 kW Induction motor and incorporated inverter switching algorithm to implement field-oriented control and maximize motor efficiency.
- Performed lap time simulations and analysis of vehicle powertrain to determine the optimal gear ratio of 5.1:1.

Academic Projects

Adaptive Cruise Control and Lane-Keep Assist of an RC vehicle

- Calibrated ultra-sonic sensors across the range of 200 – 500 mm using MATLAB polynomial fitting and applied a Kalman filter to reduce measurement noise.
- Implemented and tested a PID-controller-based adaptive cruise control on an RC vehicle. Limited traction-motor PWM commands and tuned the sampling interval to 12 ms, enabling the vehicle to maintain a 300 mm gap from obstacles.

Supervisory Controller for Series-Parallel Hybrid Powertrain

- Developed a rule-based controller in Simulink Stateflow to optimize engine and motor torque based on battery SOC, vehicle speed and driver torque demand. Implemented PID controller-based coolant flow for the cooling of the 90S20P battery pack.
- Incorporated a clutch-engagement condition tied to engine idle speed, preventing excessive clutch wear in the simulated control strategy.
- Implemented battery charge limits that varied with vehicle speed and driving mode, using SOC feedback to govern electric assistance, regenerative braking, and engine-driven charging ensuring maximum number of battery cycles.

Relevant Skills

Controls design and state-estimation: PID, MPC, state-space, Kalman filter, system modelling, CAN communication.

Simulation and system analysis: MATLAB, Simulink, Stateflow, CarSim.

Programming Languages: Python, Arduino C/C++.